

Savior Glass

An assistive smart glass for blind and low-vision users in Bangladesh, with Bangladeshi Taka recognition and counterfeit (jaal) research

Project report · 3 October 2026 (updated) · Repository branch `RPi5_deploy` (latest commit f24c93b8) · Every number below comes from a result file in the repository (claim registry: 298 claims, all pass the value check)

1. Summary

Savior Glass is a wearable camera glass built on a Raspberry Pi 5. It speaks in Bangla and English through bone conduction. It can:

- read printed text aloud;
- name objects around the user;
- recognise Bangladeshi Taka notes and give a safe genuine-or-check answer for 500 and 1,000 Taka;
- use an online vision model when the internet is available.

The research side has three parts:

- the counterfeit problem on the JaalTaka dataset;
- a watermark-based check that works on counterfeit prints the model has never seen;
- an OCR benchmark with correctly shaped Bangla text.

Main results

- **Taka detector:** mAP@0.5 0.995 on 2,282 test images, and 91.5 % correct on 1,536 independent real photos.
- **Counterfeit check, unseen prints:** 95.0 % (3 seeds, 6 views) with the watermark hybrid.
- **OCR, held-out test:** character error 9.3 % → 2.4 %; Bangla 11.9 % → 3.2 %.
- **Safety:** the glass never says "counterfeit". 0 of 88 test counterfeits were passed as genuine.
- **Raspberry Pi 5:** currency check 327 ms median; 30 minutes with no throttling (peak 77.1 °C).

Main findings

- **JaalTaka's usual split** lets the same counterfeit print appear in training and test. It is a reprint test.
- **On truly unseen prints**, accuracy drops from about 98 % to about 90 %. The banknote watermark recovers about 5 points.
- **Earlier Bangla OCR tests** used wrongly drawn Bangla (no text shaping). They were redone.
- **Remaining gains need data:** more counterfeit prints, glass-camera photos and a user study, not more models.

2. Problem statement

1. **Daily independence.** Blind and low-vision people in Bangladesh cannot read signs, labels, medicine packets or prices without help, and cannot always tell which note they are holding. Most assistive tools are English-first, need an internet connection or a phone in the hand, and do not read Bangla.
2. **Counterfeit notes.** Counterfeit 500 and 1,000 Taka notes circulate, and a blind person cannot feel a watermark. A wrong "genuine" answer costs money. A wrong "counterfeit" answer can cause a public accusation. A device must therefore be safe before it is clever.
3. **Evaluation is misleading.** Published counterfeit results on JaalTaka look near-perfect. But counterfeits printed from the same plate share one serial and appear in both training and test, so the number measures recognition of known reprints, not new counterfeits.
4. **Bangla OCR is fragile.** Bangla script has conjuncts and vowel signs that move around the consonant. Off-the-shelf OCR is weaker on Bangla than on English, and synthetic test images are easy to draw wrongly.

5. **Small hardware.** All of this has to run offline on a Raspberry Pi 5 inside a glass frame, within a few hundred milliseconds, without overheating.

3. What the glass can do (features)

Feature	How it works	Status
Text reading (OCR mode)	Capture button stores the text; the Read button speaks it. EasyOCR (Bangla + English) with OCR pipeline v2: CLAHE, tuned detector, main-text-block selection, line ordering, cleanup rules. Fallbacks: Tesseract, then the Ekush handwritten-letter CNN.	working
Object naming	YOLOv8 (COCO classes), automatic scan every 1.5 s, per-class cooldown so it does not repeat itself.	working
Currency (Taka) recognition	YOLOv8s trained on 9 Taka denominations; INT8 ONNX on the Pi; announces only above a confidence chosen on validation, to avoid naming unknown notes.	working
Safe counterfeit check	For 500 / 1,000 Taka, four views are cut from the note and scored by the PRMVT network. It says "সম্ভবত আসল" (likely genuine) only above a validation-fixed threshold, otherwise "হাতে যাচাই করুন" (check by hand). It never says "counterfeit".	working
Watermark check with guided capture	After the denomination is spoken, the glass asks the user to hold the note up to a light. It rejects dark or blurry frames with a spoken reason. A learned localizer finds the watermark window, and a 2.6 MB INT8 MobileNetV2 scores it.	on (Pi)
Online vision mode	Claude vision describes a scene when internet and an API key are available.	needs API key
Emotion-adaptive feedback	Facial-expression model (RAF-DB) can adapt feedback.	working
Speech output	Piper TTS for English (faster), espeak-ng for Bangla; queued, interruptible, volume buttons.	working
Buttons and haptics	Mode, Action, Read, Volume up / down on GPIO; vibration patterns for detection and verdict.	tested (simulated GPIO)
Fast mode switching	Mode name spoken at once; models load and warm up in the background; currency preloaded at start.	working
Study logging	Every watermark check is logged (participant, guided / unguided condition, time, rejected frames, outcome) for the user study.	ready

4. Datasets

Dataset	Content	Used for
JaalTaka	1,390 notes (802 genuine, 588 counterfeit) × 6 views; 500 and 1,000 BDT	Counterfeit research. Note-disjoint split 974 / 208 / 208 and serial-disjoint split 946 / 222 / 222 (unseen prints)
Detector composites	22,333 synthetic images (17,839 / 2,212 / 2,282) from BanglaTaka notes on backgrounds	Taka detector training and test
Fresh composite test	566 images, test notes on COCO backgrounds	Second detector test
Bangla Money (Kaggle)	1,536 real photos, 8 denominations	Independent detector test

NSTU-BDTAKA (Mendeley)	186 detection images, 1,144 close-ups	Independent detector test
Bangladeshi Counterfeit Currency Image Dataset	Whole-note photos: 1,199 genuine, 87 counterfeit (about 4 physical notes)	Whole-note counterfeit check (deployment check only)
Large Scale BDT DB 2026	100,000 images incl. coins	Surveyed; not the 9-note task
MVP-N, ModelNet40	Multi-view object sets	Test whether the view-count findings hold outside currency
COCO 2017	Everyday photos	Backgrounds, object-mode test, OCR photo scenes
RAF-DB	15,339 faces, 7 emotions	Emotion model
Ekush, MatriVasha	Bangla handwritten letters	Handwritten-letter fallback (Ekush CNN)
OCR benchmark (new)	Bangla + English phrases drawn with correct HarfBuzz shaping in 10 Bangla and 9 Latin fonts; clean, distorted and real-photo scenes; 3 seeds; separate validation and test phrases and fonts	OCR Tasks 1–7
Word lists	50k Bangla and 50k English frequency words (MIT)	OCR dictionary test, fine-tuning text

5. Models and algorithms

Model / algorithm	Role	Size / runtime
YOLOv8s Taka detector	Finds the note and names the denomination	18 MB INT8 ONNX; 254 ms median on Pi 5
Open-set confidence rule	Announce only above 0.60 (chosen on validation) to avoid naming unknown or demonetised notes	—
PRMVT (prefix-robust multi-view training)	Counterfeit network: MobileNetV3-Small + TinyViT per view, attention fusion; trained on random first-k views so it works with 1 to 6 views	16.6 MB; 36 ms on Pi 5
Prefix network, shared BN	Recommended JaalTaka model (chosen on validation)	—
Policy E (safe threshold)	"Likely genuine" only above the highest validation counterfeit score; otherwise "check by hand". Backed by Theorem 8 (pass probability $1/(n+1)$)	—
Image-quality gate	Rejects dark, blurred or occluded frames before any verdict	—
Watermark window detector	SIFT + RANSAC registration to a note template, crop of the watermark window	372 ms on Pi 5
Learned watermark localizer	MobileNetV3-Small regresses the window's four corners; no template, no SIFT	4.3 MB; 50 ms with classifier on Pi 5
Watermark classifier	MobileNetV2, static INT8 (same decisions as FP32)	2.6 MB
Hybrid	Logistic regression fitted on validation over network score + watermark score	—
Fine-tuned ResNet-50	Fair image baseline on full-resolution views	—
Capture guide	Brightness and Laplacian-sharpness thresholds (2nd percentile of validation photos), spoken prompts	—
EasyOCR + OCR pipeline v2	CRAFT text detector + CRNN recognizer (Bangla + English); dominant-text-block selection; CLAHE; tuned thresholds; cleanup rules	~1.9 s per frame on laptop CPU
Tesseract 5, Ekush CNN	OCR fallbacks (printed / handwritten letters)	Ekush 0.45 MB
YOLOv8n (COCO)	Object naming	207 ms on Pi 5
Emotion CNN	RAF-DB facial expression	48 ms on Pi 5
Piper / espeak-ng	Speech	English 3.3 s (Piper) per sentence on Pi 5
Statistics	Exact McNemar, Wilson intervals, three seeds, noncentral-t power analysis, Holm correction	—

6. Results

6.1 Taka denomination detector

Test	Result
Synthetic composites, 2,282 images	Precision 0.995, recall 0.996, mAP@0.5 0.995, mAP@0.5:0.95 0.849
Fresh composites, 566 images	mAP@0.5 0.992

Bangla Money, 1,536 real photos (no retraining)	91.5 % correct denomination
NSTU-BDTAKA detection, 186 images	AP50 0.796
NSTU hand-held close-ups, 1,144	18.5 % (very different framing; a known weakness)
Open-set rule (Bangla Money test)	1-taka notes named as another value: 43.6 % → 22.8 %

6.2 Counterfeit research on JaalTaka (note-disjoint split, 208 test notes, 3 seeds)

Method	1 view	6 views
Fixed 6-view training, shared BN	71.2 ± 11.4 %	98.1 ± 0.5 %
Prefix training, shared BN (recommended)	98.2 ± 0.7 %	98.7 ± 0.3 %
PRMVT (prefix, per-count BN)	96.5 ± 1.5 %	98.1 ± 1.0 %
CNN+ViT baseline	76.1 ± 3.6 %	92.3 ± 0.8 %
Frozen ResNet-50 probe	98.1 %	99.0 %

Findings:

- **Prefix training** removes the one-view collapse of fixed-view training (McNemar p down to 10^{-22}) and makes results stable across seeds.
- **Frozen ImageNet backbones match it on this split:** 60 paired tests, none significant. So no state-of-the-art claim is made on JaalTaka.

6.3 Unseen counterfeit prints (serial-disjoint split, 222 test notes, 101 counterfeit)

Method (3 seeds)	1 view	6 views
Frozen ResNet-50 probe	85.1 %	83.8 %
Prefix network	89.9 ± 0.7 %	89.3 ± 1.3 %
Fine-tuned ResNet-50, full resolution (fair baseline)	94.9 ± 1.8 %	94.4 ± 2.6 %
Prefix network + watermark (hybrid)	94.4 ± 0.5 %	95.0 ± 0.0 %
Watermark alone, MobileNetV2 (registered notes)	92.9 %, AUC 0.976	
Watermark with learned localizer (all 222 notes)	92.6 ± 0.5 % (ties SIFT, p = 1.0; covers 222 / 222 instead of 197)	

Findings:

- Accuracy drops by about 8 points when counterfeit prints are truly new.
- The watermark brings it back to 95 % and is stable across seeds.
- The hybrid ties a properly fine-tuned ResNet-50. It does not significantly beat it: there are only about 24 distinct test prints.
- A serial blacklist catches 0 of 19 unseen-serial counterfeits.

6.4 Safety of the deployed counterfeit check (policy E)

Measure	Result
JaalTaka test counterfeits passed as "likely genuine"	0 / 88 (95 % CI 0–4.2 %)
Whole-note counterfeit photos passed (app code path)	0 / 20
"Counterfeit" spoken in 1,889 app runs	0 times

Wrong verdicts under bad light, with the quality gate	0 / 208 per condition
Cost: genuine whole notes confirmed	low (0.4–2 % on independent photos): the glass often asks for a hand check

6.5 OCR (new benchmark, held-out test, read once per method)

Pipeline	Overall CER	Bangla	English	Signs in photos
Glass app before this work	9.3 %	11.9 %	6.6 %	10.7 %
OCR pipeline v2 (now in the app)	2.4 %	3.2 %	1.6 %	2.2 %
Tesseract 5 on the text block (best other engine)	16.6 %	18.5 %	14.7 %	32.5 %
PaddleOCR (English only; no Bangla model)	11.0 %	—	11.0 %	33.1 %

What was tested (Tasks 1–7):

- **Kept:**
 - text-region selection, the largest gain;
 - CLAHE (a marginal gain);
 - EasyOCR parameter tuning (31 settings);
 - cleanup rules.
- **Not kept:**
 - binarisation;
 - multi-scale reading;
 - dictionary and ByT5 correction (ByT5 invents text);
 - the old lexicon repair;
 - TrOCR;
 - fine-tuning the recognizer: better Bangla, worse English on validation. It was better on test, but it was not adopted, because adopting it would mean choosing on the test set.
- **Speed:** v2 costs no extra time.

6.6 Other assistive modules

Module	Result
Emotion, RAF-DB test (3,068 faces)	86.5 % accuracy, macro-F1 0.795
Object detection, COCO val2017 subset	mAP@0.5 0.604
Ekush handwritten letters (122 classes, unseen writers)	90.0 %
Guided capture thresholds (test back-lit photos)	213 / 222 clean accepted; 0 / 222 darkened or defocused accepted
Mode switch, first currency check (laptop CPU)	661 ms → 43 ms after warm-up

6.7 Raspberry Pi 5 (measured 1 Oct 2026, 8 GB, 100 iterations + 30-minute run)

Module	Median	95th percentile
Currency mode, end to end (detect + check)	327 ms	560 ms
Taka detector, INT8 ONNX	254 ms	339 ms
Watermark check, learned localizer	50 ms	62 ms
Watermark check, SIFT path	372 ms	854 ms
Counterfeit network (QDUIG / PRMVT)	36 ms	50 ms
Object mode (YOLOv8n)	207 ms	217 ms
Emotion	48 ms	52 ms

OCR mode (EasyOCR, canvas 2560)	18.7 s	20.4 s
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System behaviour during the run:

- Peak memory 1.48 GB.
- Peak temperature 77.1 °C, with no throttling in 30 minutes (1,229 calls).
- Currency from button press to finished speech: about 2.3–2.9 s.
- OCR on the Pi is slow (about 19 s). Canvas 640 is about 3× faster, for about one point more Bangla error.

6.8 Theory

- **Theorem 8 (safe threshold).** With the threshold set at the highest of n validation counterfeit scores, the probability that a new counterfeit passes is $1/(n+1)$. For $n = 88$, that is 1.1 %.
- **Theorem 9.** Under a distribution shift the pass rate grows by at most the total-variation distance.
- **Theorem 10.** The quality gate's false-rejection bound.
- **Proposition 11.** Fixed-view training does not identify the few-view predictor, which explains the seed instability.
- **Theorem 16.** The effective test size is the number of prints, not notes.

7. What is done and what is not

Item	Status	Note
Glass app: OCR, objects, currency, safe check, watermark + guided capture, speech, buttons, haptics	done	All glass tests pass on the laptop
Taka detector, cross-dataset tests, INT8 export	done	
JaalTaka research: 18 methods × 3 seeds, serial audit, unseen-print split, watermark hybrid, fair ResNet baseline	done	
Learned watermark localizer, print-locked capture tool, study logging and analysis script	done	
OCR Tasks 1–7 and pipeline v2 in the app	done	On a synthetic benchmark
Raspberry Pi 5 benchmark and 30-minute thermal run	done	OCR v2 and mode-switch speed not yet re-measured on the Pi
Papers (workshop, D&B, journal drafts), theorems, figures, claim registry	done	Author names and affiliation still missing
Whole-note counterfeit photos from the glass camera, grouped by physical note	not done	Needed to trust a whole-note counterfeit verdict
Print-disjoint counterfeit dataset (~100 unseen prints) with a bank or police partner	not done	Tools ready; data collection needed
Second currency (INR or USD)	not done	Breaks the earlier "Taka only" rule; your decision
User study with blind / low-vision participants	not done	Protocol, pre-registration and analysis ready
Glass-camera calibration of the capture-guide thresholds	not done	Current values come from JaalTaka photos
Battery life	not done	Needs a USB-C power meter
Bangla Piper voice on the Pi	blocked	The ARM Piper build cannot phonemise the Bangla voice; falls back to espeak-ng
Speech speed (135 vs 175 words per minute)	your decision	Can be tried with ESPEAK_SPEED=175
OCR on real glass photos; fine-tuning with real data	not done	Current OCR numbers are synthetic
KenLM, DBNet (OpenCV), EAST, YOLO text detector	done	All run on 3 Oct. None beats the current detector or the cleanup rules, so nothing changed in the app
Online vision mode measured	blocked	OpenAI support added and an evaluation script is ready; the supplied account has no credits (HTTP 429), so no number exists
Live-camera note accuracy on the glass	not done	Needs glass photos
Revoke the Pi's GitHub deploy key	your action	GitHub → Settings → Deploy keys

8. Limitations

- **Counterfeit results rest on about 24 distinct counterfeit prints** in the unseen-print test. The uncertainty is about ± 12 points at the print level.
- **Whole-note counterfeit checking on the glass is not proven.** The model was trained on close-ups, so the glass is set to be safe rather than decisive.
- **OCR numbers come from rendered text.** Real signs, glare and curved packets will be harder.
- **Detector accuracy drops on hand-held close-ups** (NSTU, 18.5 %).
- **No user has worn the glass yet** in a study.

9. Research questions

1. **RQ1.** How much does counterfeit-detection accuracy fall when every test counterfeit comes from a print never seen in training, and what is the right unit of evaluation: notes, serials or prints?
2. **RQ2.** Can a physical security feature (the transmitted-light watermark), found by a learned localizer, generalise to unseen prints and to a second currency better than a fine-tuned whole-note CNN?
3. **RQ3.** Does training on random prefixes of views (1 to k) remove the accuracy collapse that fixed-view multi-view models show when fewer views are available, and does this hold beyond banknotes (MVP-N, ModelNet)?
4. **RQ4.** Can an assistive device give provably safe counterfeit feedback, with zero "counterfeit" accusations and a bounded chance of passing a counterfeit, while still being useful to blind users?
5. **RQ5.** Does guided capture (spoken prompts to hold the note to the light, rejecting dark or blurry frames) reduce the time per note and the number of "check by hand" answers for blind and low-vision users?
6. **RQ6.** Which parts of a Bangla OCR pipeline matter most for scene text read by a wearable camera: detection and region selection, preprocessing, recognizer fine-tuning, or language post-processing?
7. **RQ7.** How far can a multi-function assistive system (OCR, objects, currency, counterfeit check) be pushed on a Raspberry Pi 5 within latency, memory and thermal limits that a wearable needs?
8. **RQ8.** How large must a print-disjoint counterfeit dataset be, in prints, to show a significant difference between two detectors?

Sources in the repository:

- `paper_evidence/FINAL_RESULTS.md`, `BEAT_RESNET50_FINAL.md`, `WATERMARK_LOCALIZER.md`, `PI5_RESULTS.md`
- `OCR_*.md`, `JAAL_VERDICT_FIXED.md`, `THEOREMS.md`, `A_STAR_ROADMAP.md`
- `CLAIM_REGISTRY.json`